

# Appointment Details With Patient, Doctor, And Department

## Database Description

The hospital\_mgmt database is designed to efficiently organize and manage healthcare operations. It maintains structured records of hospital departments, doctors, patients, and appointment schedules. This system enables seamless tracking of doctor-patient interactions, appointment statuses, and departmental medical activities, supporting both clinical insights and administrative decision-making.

## ✓ Schema Specification

### Table: Departments

Stores hospital departments like Cardiology, Neurology, etc.

| Column Name   | Data Type    | Constraints | Description                      |
|---------------|--------------|-------------|----------------------------------|
| department_id | INT          | PRIMARY KEY | Unique identifier for department |
| name          | VARCHAR(100) | NOT NULL    | Name of the department           |

### Table: Doctors

Stores basic doctor information including their assigned department.

| Column Name    | Data Type    | Constraints                                 | Description              |
|----------------|--------------|---------------------------------------------|--------------------------|
| doctor_id      | INT          | PRIMARY KEY                                 | Unique ID for the doctor |
| name           | VARCHAR(100) | NOT NULL                                    | Full name                |
| specialization | VARCHAR(100) |                                             | Medical specialty        |
| department_id  | INT          | FOREIGN KEY →<br>Departments(department_id) | Department they work in  |

### Table: Patients

Stores registered patients and their demographics.

| Column Name | Data Type    | Constraints | Description                   |
|-------------|--------------|-------------|-------------------------------|
| patient_id  | INT          | PRIMARY KEY | Unique identifier for patient |
| name        | VARCHAR(100) | NOT NULL    | Full name of the patient      |
| gender      | VARCHAR(10)  |             | Gender                        |
| age         | INT          |             | Age                           |
| phone       | VARCHAR(15)  |             | Contact number                |

### Table: Appointments

Stores patient appointments with doctors.

| Column Name      | Data Type   | Constraints                        | Description                     |
|------------------|-------------|------------------------------------|---------------------------------|
| appointment_id   | INT         | PRIMARY KEY                        | Unique appointment identifier   |
| patient_id       | INT         | FOREIGN KEY → Patients(patient_id) | Patient who has the appointment |
| doctor_id        | INT         | FOREIGN KEY → Doctors(doctor_id)   | Doctor consulted                |
| appointment_date | DATE        | NOT NULL                           | Date of the appointment         |
| status           | VARCHAR(20) | DEFAULT 'Scheduled'                | Appointment status              |

## ✓ Test Data

### Departments

| department_id | name        |
|---------------|-------------|
| 1             | Cardiology  |
| 2             | Neurology   |
| 3             | Pediatrics  |
| 4             | Orthopedics |

### Doctors

| doctor_id | name       | specialization     | department_id |
|-----------|------------|--------------------|---------------|
| 501       | Dr. Mehta  | Cardiologist       | 1             |
| 502       | Dr. Rao    | Neurologist        | 2             |
| 503       | Dr. Sharma | Pediatrician       | 3             |
| 504       | Dr. Nair   | Orthopedic Surgeon | 4             |

### Patients

| patient_id | name           | gender | age | phone      |
|------------|----------------|--------|-----|------------|
| 601        | Ananya Verma   | Female | 30  | 9876543210 |
| 602        | Rajesh Khanna  | Male   | 45  | 8765432109 |
| 603        | Priya Desai    | Female | 27  | 7654321098 |
| 604        | Manish Agarwal | Male   | 38  | 6543210987 |

### Appointments

| appointment_id | patient_id | doctor_id | appointment_date | status    |
|----------------|------------|-----------|------------------|-----------|
| 701            | 601        | 501       | 2023-08-01       | Completed |
| 702            | 602        | 502       | 2023-08-03       | Cancelled |
| 703            | 603        | 503       | 2023-08-04       | Completed |
| 704            | 604        | 504       | 2023-08-05       | Scheduled |
| 705            | 601        | 502       | 2023-08-06       | Completed |
| 706            | 602        | 503       | 2023-08-07       | Completed |
| 707            | 603        | 504       | 2023-08-08       | Completed |
| 708            | 604        | 501       | 2023-08-09       | Scheduled |
| 709            | 601        | 503       | 2023-08-10       | Completed |
| 710            | 602        | 504       | 2023-08-11       | Completed |
| 711            | 603        | 501       | 2023-08-12       | Cancelled |
| 712            | 604        | 502       | 2023-08-13       | Completed |

### Problem Statement

Generate a report listing all appointments along with patient name, doctor name, and the department the doctor belongs to. This report will be used for internal review and activity tracking.

### Query to be Implemented

Write a SQL query to fetch:

- appointment\_id
- appointment\_date
- status
- patient\_name
- doctor\_name

- department\_name

### Expected Output (Screenshots are Continued)

| appointment_id | appointment_date         | status    | patient_name   | doctor_name | department_name |
|----------------|--------------------------|-----------|----------------|-------------|-----------------|
| 701            | 2023-08-01T00:00:00.000Z | Completed | Ananya Verma   | Dr. Mehta   | Cardiology      |
| 702            | 2023-08-03T00:00:00.000Z | Cancelled | Rajesh Khanna  | Dr. Rao     | Neurology       |
| 703            | 2023-08-04T00:00:00.000Z | Completed | Priya Desai    | Dr. Sharma  | Pediatrics      |
| 704            | 2023-08-05T00:00:00.000Z | Scheduled | Manish Agarwal | Dr. Nair    | Orthopedics     |

| appointment_id | appointment_date         | status    | patient_name   | doctor_name | department_name |
|----------------|--------------------------|-----------|----------------|-------------|-----------------|
| 705            | 2023-08-06T00:00:00.000Z | Completed | Ananya Verma   | Dr. Rao     | Neurology       |
| 706            | 2023-08-07T00:00:00.000Z | Completed | Rajesh Khanna  | Dr. Sharma  | Pediatrics      |
| 707            | 2023-08-08T00:00:00.000Z | Completed | Priya Desai    | Dr. Nair    | Orthopedics     |
| 708            | 2023-08-09T00:00:00.000Z | Scheduled | Manish Agarwal | Dr. Mehta   | Cardiology      |
| 709            | 2023-08-10T00:00:00.000Z | Completed | Ananya Verma   | Dr. Sharma  | Pediatrics      |

|     |                          |           |                |           |             |
|-----|--------------------------|-----------|----------------|-----------|-------------|
| 710 | 2023-08-11T00:00:00.000Z | Completed | Rajesh Khanna  | Dr. Nair  | Orthopedics |
| 711 | 2023-08-12T00:00:00.000Z | Cancelled | Priya Desai    | Dr. Mehta | Cardiology  |
| 712 | 2023-08-13T00:00:00.000Z | Completed | Manish Agarwal | Dr. Rao   | Neurology   |